Vikash Vashisth

CFD Engineer | Thermal-Fluid and Multiphysics Simulation

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PROFESSIONAL SUMMARY

CFD engineer with a PhD in Chemical Engineering and experience in academic and industry-linked R&D, specialising in thermal-fluid analysis, reacting and multiphase flows, turbulence modelling and heat transfer. Combines numerical verification, mesh-independence assessment and experimental validation to evaluate equipment performance and inform design and operating decisions. Advanced ANSYS Fluent expertise, complemented by STAR-CCM+, electromagnetic-CFD analysis and Python-based simulation automation.

CORE EXPERTISE

Physics: Reacting flows; Plasma CFD and finite-rate chemistry; multiphase; heat and mass transfer; thermal management; species transport; rotating machinery and non-Newtonian flows.

Methods: RANS/LES; steady and transient CFD; sliding-mesh; adaptive mesh refinement; mesh-independence assessment; convergence and conservation; experimental validation; HPC.

Software: ANSYS Fluent; STAR-CCM+; ANSYS HFSS; Python, PyFluent and PyAEDT; MATLAB; UDF/C; Siemens NX; Workbench, Meshing, DesignModeler; Linux.

PROFESSIONAL EXPERIENCE

Chalmers University of Technology — Gothenburg, Sweden

Researcher - CFD and Multiphysics — Dec 2025 - Present

Postdoctoral Researcher - CFD and Multiphysics — Dec 2022 - Dec 2025

- Modelled and experimentally validated microwave plasma reactors using 3D CFD, incorporating fluid flow, heat transfer, species transport and finite-rate chemistry.
- Quantified reactor performance, predicting 6-10% higher NO yield and 6-8% lower specific energy consumption relative to reference configurations.
- Developed fully automated two-way HFSS-Fluent coupling using PyAEDT and PyFluent, integrating electromagnetic power deposition with fluid flow, heat transfer and finite-rate chemistry.
- Simulated multiphase bubble and droplet breakup using LES-VOF and adaptive mesh refinement, predicting interface dynamics and fragment-size distributions.
- Verified LES resolution and numerical sensitivity, and validated predictions against high-speed experimental measurements.
- Led biweekly CFD reviews with NitroCapt AB and research partners, presenting simulation results and engineering recommendations.

PROFESSIONAL EXPERIENCE - CONTINUED

IIT Guwahati - Reckitt Benckiser Collaboration Guwahati, India

Research Associate - Industrial CFD Project Jan 2022 - Apr 2022

- Simulated shear-thinning liquid spreading using transient 3D VOF with experimentally measured rheology.
- Established mesh-independent predictions and validated liquid-spreading behaviour against experimental measurements.

IIT Roorkee Roorkee, India

Doctoral Researcher - Fluid Mechanics, Rotating Machinery and Mixing Dec 2015 - Apr 2021

- Analysed rotor-stator mixer performance using sliding-mesh RANS and LES, evaluating mesh-independence, flow distribution, mixing efficiency and power consumption across mixer designs.
- Compared conventional and newly developed configurations, with predicted power numbers agreeing within approximately 10% of benchmark measurements.

SELECTED ENGINEERING PROJECT

Electric Drive Unit Cooling Jacket - Conjugate Heat Transfer 2026

Independent engineering simulation and geometry-comparison study

- Built a liquid-cooled electric drive unit CHT model in STAR-CCM+, coupling coolant flow with heat conduction through the housing and equivalent stator.
- Established mesh-independent CFD predictions and evaluated cooling-channel geometry variations.
- Quantified the cooling-versus-pressure-loss trade-off: seven turns reduced peak equivalent-stator temperature by 1.4 K while increasing pressure loss by 16.2% relative to the six-turn reference case.

EDUCATION

Indian Institute of Technology Roorkee Roorkee, India

PhD, Chemical Engineering 2015 - 2021

Thesis: Hydrodynamics and Mixing Studies in High Shear Mixers for Viscous Fluids

M.Tech, Chemical Engineering 2011 - 2013

SELECTED PUBLICATIONS

- Analysis of forced and intrinsic quenching in a plasma reactor for nitrogen fixation by use of three-dimensional simulations and experiments. Chemical Engineering Journal (2026).
- Dynamics of bubble breakup under turbulent flow conditions. Chemical Engineering Journal (2025).
- Analysis of a plasma reactor performance for direct nitrogen fixation by use of three-dimensional simulations and experiments. Chemical Engineering Journal (2024).

TRAINING & ADDITIONAL INFORMATION

Training: Ansys Learning Hub - PyFluent and HFSS (2026).

Languages and mobility: English - fluent; Swedish - A2 certificate, continuing studies; Swedish driving licence (B); open to relocation.